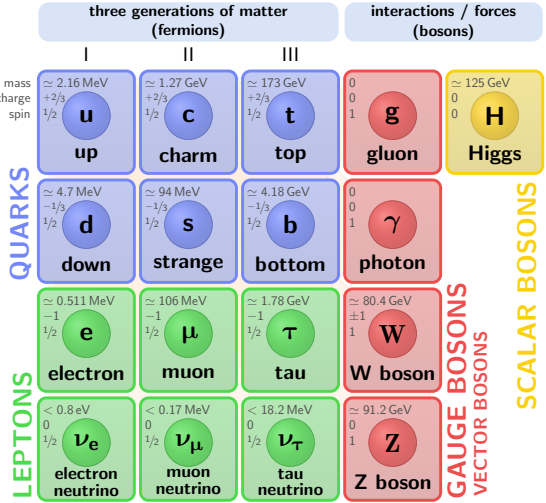
# Standard Model of Particle Physics

Izaak Neutelings

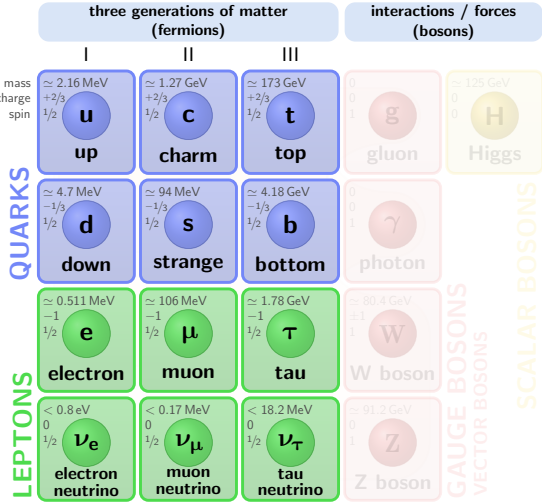
University of Zurich

2023

# Standard Model of Particle Physics

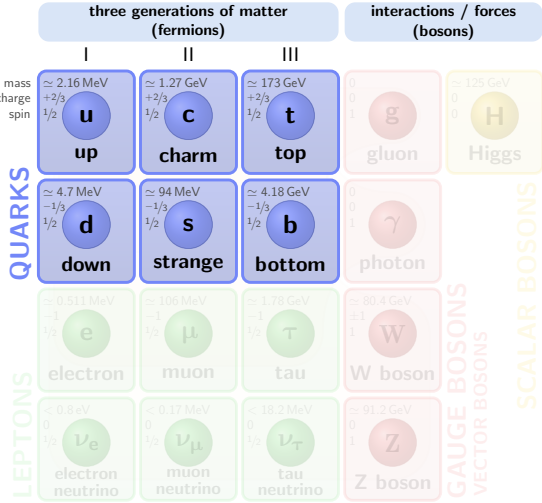


# Standard Model of Particle Physics



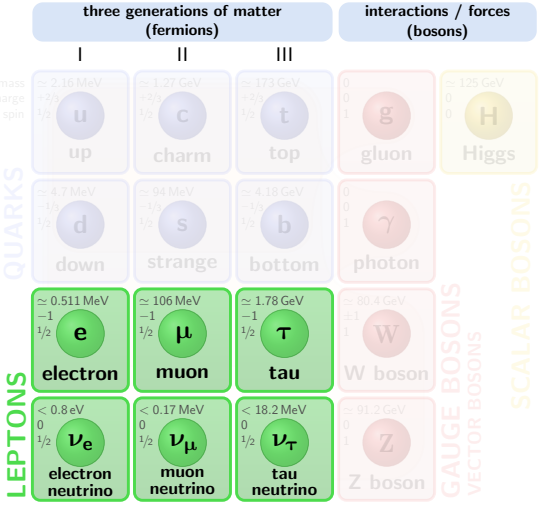

- **Fermions** of spin 1/2, which make up matter

# Standard Model of Particle Physics



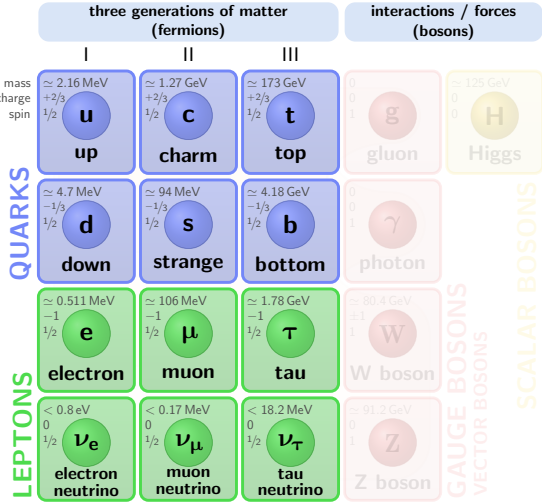

- **Fermions** of spin 1/2, which make up matter
  - **Quarks**

# Standard Model of Particle Physics



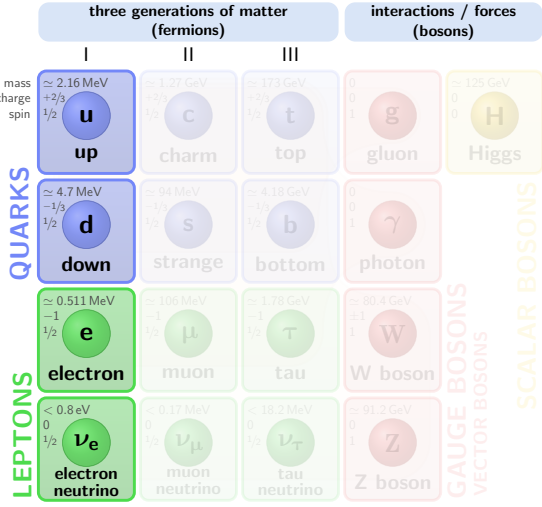

- **Fermions** of spin 1/2, which make up matter
  - **Quarks**
  - **Leptons**

# Standard Model of Particle Physics



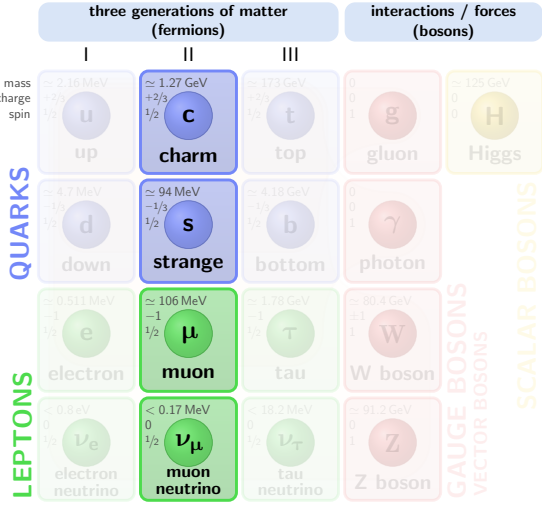

- **Fermions** of spin 1/2, which make up matter
  - **Quarks**
  - **Leptons**
  - Three generations

# Standard Model of Particle Physics



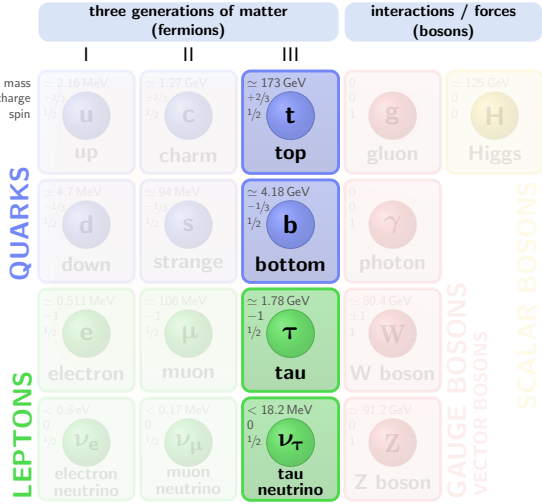

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# Standard Model of Particle Physics



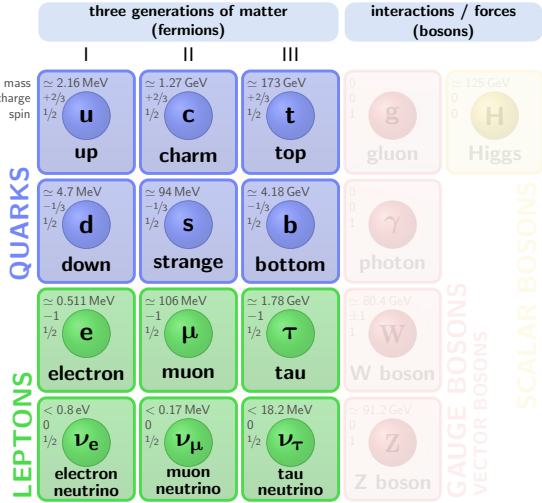

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# Standard Model of Particle Physics



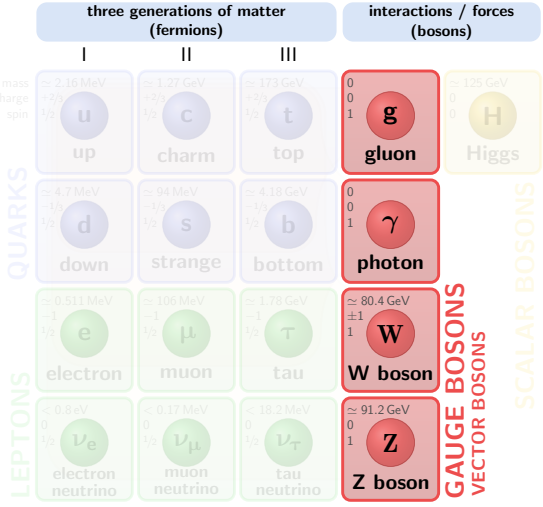

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# Standard Model of Particle Physics



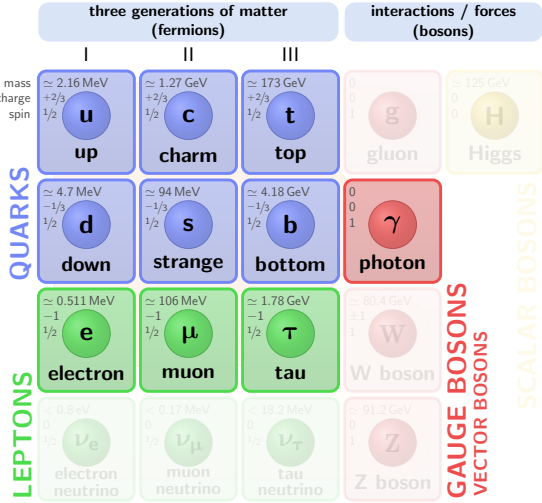

- **Fermions** of spin 1/2, which make up matter
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# Standard Model of Particle Physics



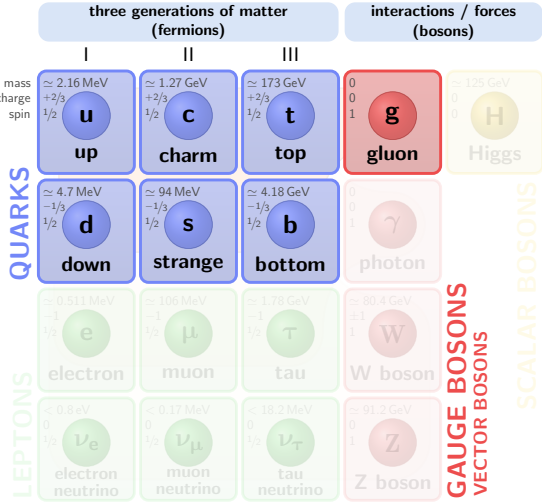

- **Fermions** of spin 1/2, which make up matter
  - **Quarks**
  - **Leptons**
  - Three generations
- **Gauge bosons** of spin 1, which mediate the fundamental interactions

# Standard Model of Particle Physics



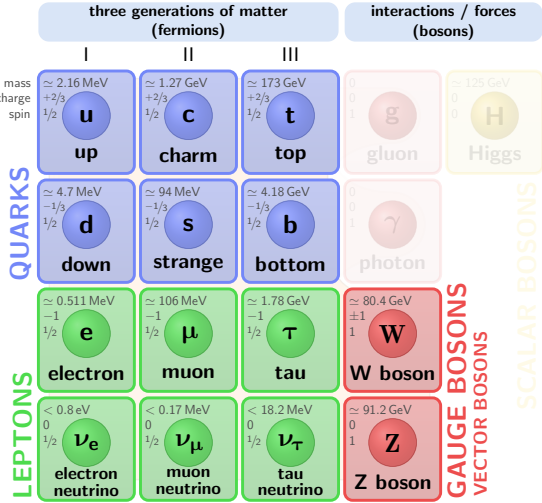

- **Fermions** of spin 1/2, which make up matter
  - **Quarks**
  - **Leptons**
  - Three generations
- **Gauge bosons** of spin 1, which mediate the fundamental interactions
  - Electromagnetic interaction

# Standard Model of Particle Physics



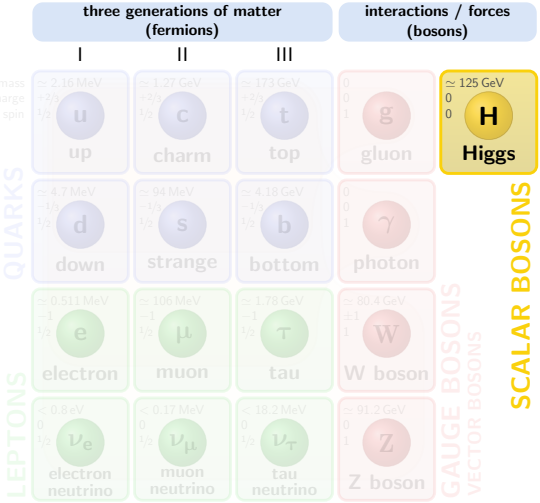

- **Fermions** of spin 1/2, which make up matter
  - **Quarks**
  - **Leptons**
  - Three generations
- **Gauge bosons** of spin 1, which mediate the fundamental interactions
  - Electromagnetic interaction
  - Strong interaction

# Standard Model of Particle Physics



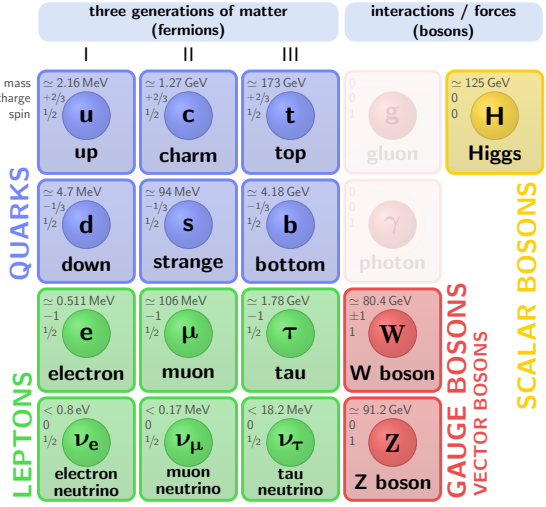

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  - **Quarks**
  - **Leptons**
  - Three generations
- **Gauge bosons** of spin 1, which mediate the fundamental interactions
  - Electromagnetic interaction
  - Strong interaction
  - Weak interaction

# Standard Model of Particle Physics



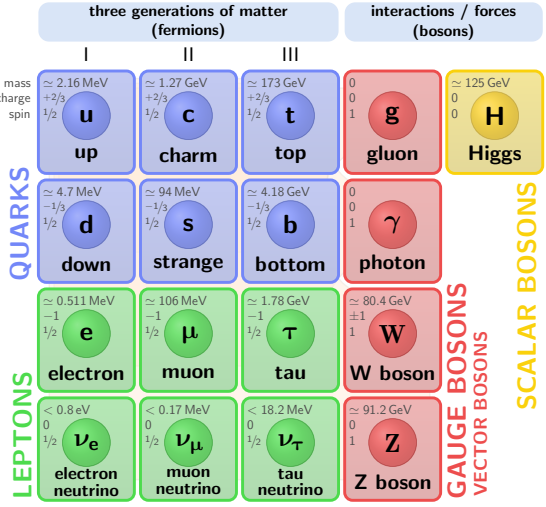

- **Fermions** of spin 1/2, which make up matter
  - **Quarks**
  - **Leptons**
  - Three generations
- **Gauge bosons** of spin 1, which mediate the fundamental interactions
  - Electromagnetic interaction
  - Strong interaction
  - Weak interaction
- One **scalar boson** of spin 0, which gives mass to fermions and weak bosons

# Standard Model of Particle Physics




- **Fermions** of spin 1/2, which make up matter
  - **Quarks**
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  - Three generations
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# Standard Model of Particle Physics




- **Fermions** of spin 1/2, which make up matter
  - **Quarks**
  - **Leptons**
  - Three generations
- **Gauge bosons** of spin 1, which mediate the fundamental interactions
  - Electromagnetic interaction
  - Strong interaction
  - Weak interaction
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